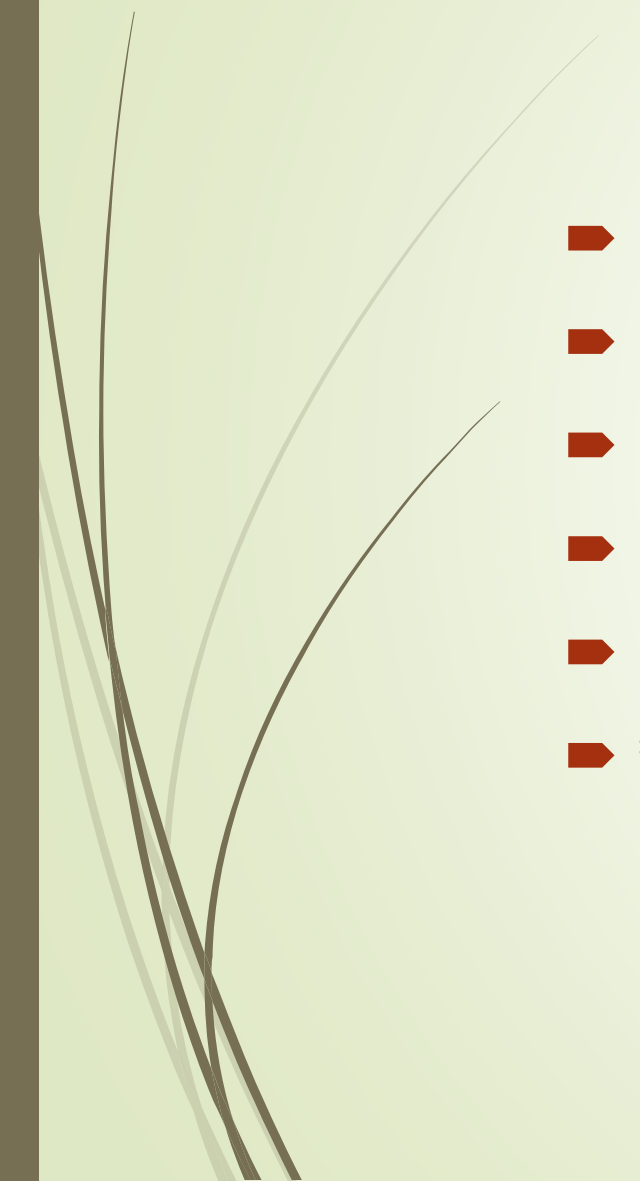




Chapter 3 Multivariate Normal Distribution



Content



- § 3.1 Definition of Multivariate Normal Distribution (omit)
- § 3.2 Properties of Multivariate Normal Distribution (omit)
- § 3.3 Multiple and partial correlation coefficient
- § 3.4 Maximum Likelihood Estimation and properties of estimators
- § 3.5 Sample Distribution about $\bar{\mathbf{x}}$ and $(n - 1) S$
- * § 3.6 Quadric Form Distribution

§ 3.1 Definition of Multivariate Normal Distribution

- The probability density function (pdf) of univariate normal distribution $N(\mu, \sigma^2)$ is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$
$$= (2\pi)^{-1/2} (\sigma^2)^{-1/2} \exp\left[-\frac{1}{2}(x-\mu)(\sigma^2)^{-1}(x-\mu)\right], \quad -\infty < x < \infty$$

- If the pdf of random vector $\mathbf{x} = (x_1, x_2, \dots, x_p)'$ is

$$f(\mathbf{x}) = (2\pi)^{-p/2} |\boldsymbol{\Sigma}|^{-1/2} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right],$$

then $\mathbf{x}$ obeys p -variate normal distribution (p元正态分布) denoted by $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where parameters $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the mean and covariance matrix of $\mathbf{x}$ such that $\boldsymbol{\Sigma} > \mathbf{0}$.

Example 3.1.1 (bivariate normal distribution)

➤ Let $\mathbf{x} \sim N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

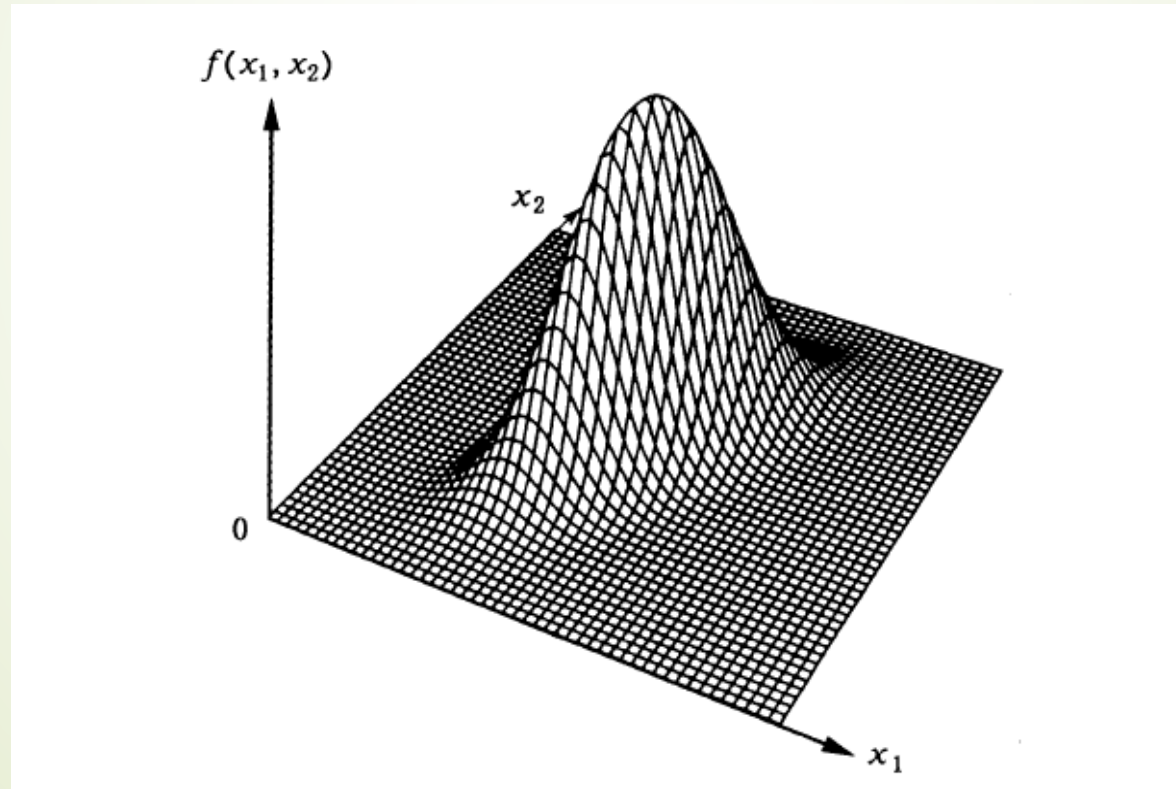
$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_1\sigma_2\rho \\ \sigma_1\sigma_2\rho & \sigma_2^2 \end{pmatrix},$$
$$\boldsymbol{\Sigma}^{-1} = \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)} \begin{bmatrix} \sigma_2^2 & -\sigma_1\sigma_2\rho \\ -\sigma_1\sigma_2\rho & \sigma_1^2 \end{bmatrix}$$

obviously ρ is the correlation coefficient of x_1 and x_2 . When $|\rho| < 1$, the probability density function of $\mathbf{x}$ is

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \cdot \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x_1 - \mu_1}{\sigma_1} \right) \left(\frac{x_2 - \mu_2}{\sigma_2} \right) + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right] \right\}$$

Density Surface Diagram of Bivariate Normal Distribution

- When $\sigma_1^2 = \sigma_2^2, \rho = 0.75$, the bell shaped density surface diagram of bivariate normal distribution is shown below:



Density Surface Diagram of Bivariate Normal Distribution

- When $\mu_1 = \mu_2 = 10$, $\sigma_1^2 = \sigma_2^2 = 3$, $\rho = 0.75$, the following matlab codes can be used to produce the bell shaped density surface diagram of bivariate normal distribution.

Matlab codes:

```
u1 = 10; %均值
```

```
u2 = 10;
```

```
sigma1 = 3; %方差
```

```
sigma2 = 3;
```

```
rou = 0.75; %相关系数
```

```
x=0:0.5:20;
```

```
y=0:0.5:20;
```

```
[X,Y]=meshgrid(x,y); %基于向量x和y中包含的坐标返回二维网格坐标
```

```
p = 1/(2*pi*sigma1*sigma2*sqrt(1-rou*rou)).*exp(-1/(2*(1-rou^2)).*[(X-u1).*(X-u1)/(sigma1*sigma1)-  
2*rou*(X-u1).*(Y-u2)/(sigma1*sigma2)+(Y-u2).*(Y-u2)/(sigma2*sigma2)]);
```

```
mesh(X,Y,p); %用来绘制三维网格图
```

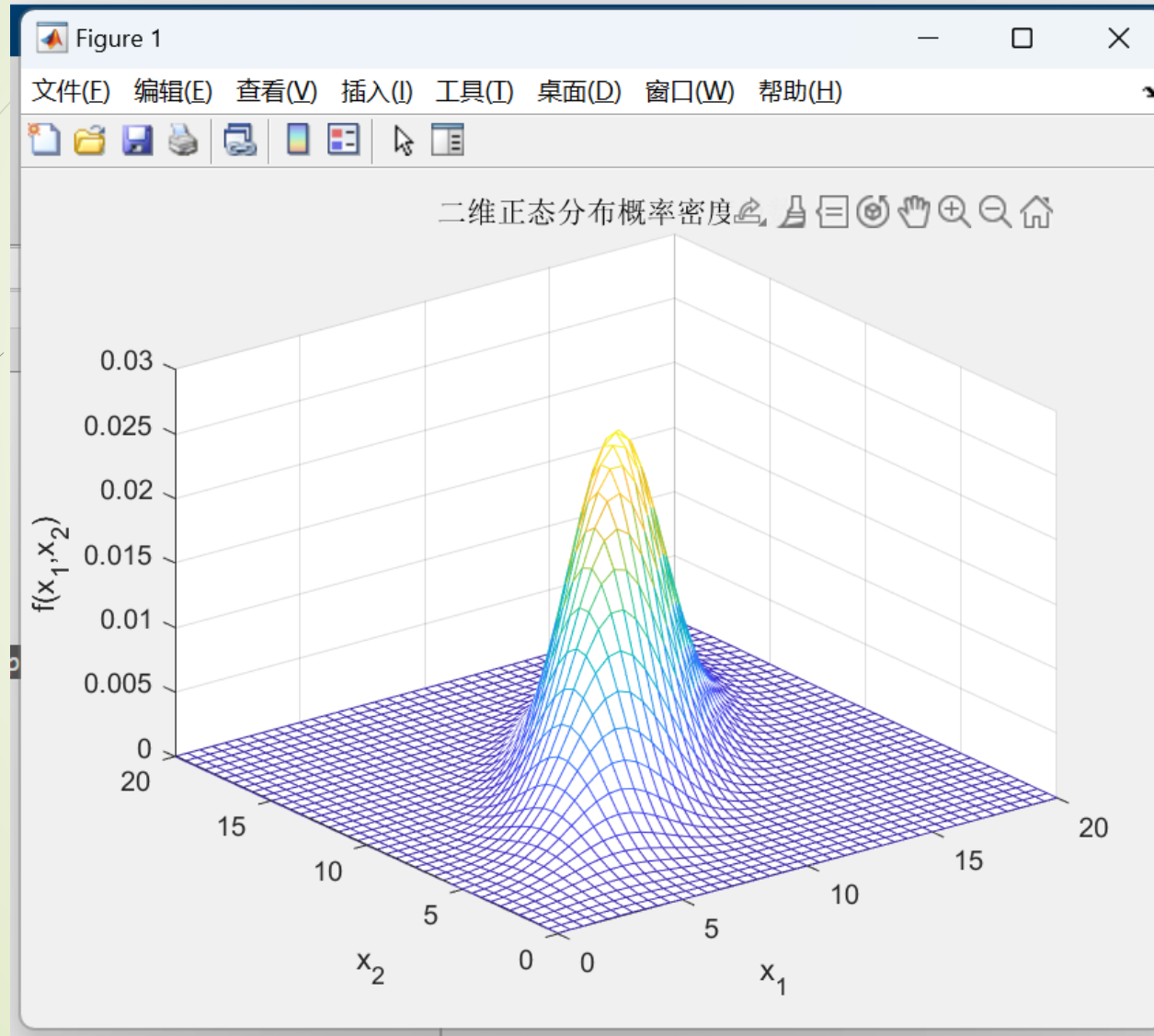
```
xlabel('x_1');
```

```
ylabel('x_2');
```

```
zlabel('f(x_1,x_2)');
```

```
title('二维正态分布概率密度函数');
```


The bivariate normal pdf with $\mu_1 = \mu_2 = 10$, $\sigma_1^2 = \sigma_2^2 = 3$, $\rho = 0.75$



Isohypse Line(等高线) of Bivariate Normal Distribution

➤ Contour (ellipse) Line:

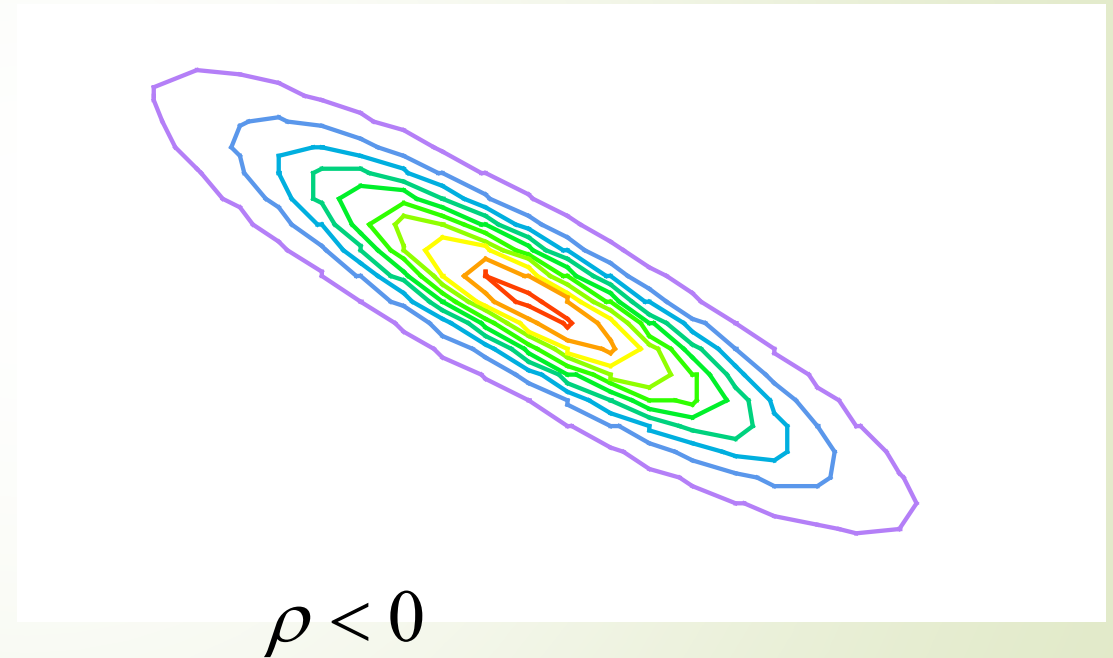
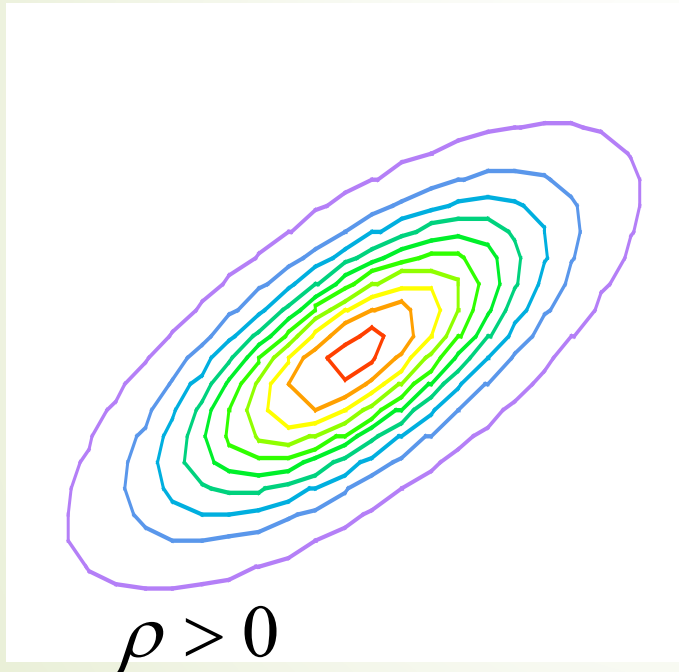
$$\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1 - \mu_1}{\sigma_1}\right)\left(\frac{x_2 - \mu_2}{\sigma_2}\right) + \left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2 = c^2$$

➤ Density of the above-mentioned contour line is

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{c^2}{2(1-\rho^2)}\right]$$

Contour Line Group of Bivariate Normal Distribution

(Generated by 10,000 random numbers from SAS/Insight)



§ 3.2 Properties of Multivariate Normal Distribution

- ➡ * (1) omitted.
- ➡ (2) Let $\mathbf{x}$ be a p dimensional random vector, then $\mathbf{x}$ obeys multivariate normal distribution, if and only if any of its linear functions $\mathbf{a}'\mathbf{x}$ obey single normal distribution.
- Property(2) can be used to prove that random vectors obey multivariate normal distribution.
- ➡ (3) Let $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{b}$, where $\mathbf{C}$ is $r \times p$ constant matrix, then
$$\mathbf{y} \sim N_r(\mathbf{C}\boldsymbol{\mu} + \mathbf{b}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}')$$
- The property shows that any linear transformation of a (multivariate) normal variable is still (multivariate) normal variable.

Exercise: P 69 3.3

- 
- **Example 3.2.2** Let $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{a}$ be a p dimensional constant vector, then from the above properties (2) or (3), we can get

$$\mathbf{a}'\mathbf{x} \sim N(\mathbf{a}'\boldsymbol{\mu}, \mathbf{a}'\boldsymbol{\Sigma}\mathbf{a})$$

- (4) Let $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then any sub-vector of $\mathbf{x}$ obey (multivariate) normal distribution. Its mean (covariance matrix) is the corresponding sub-vector (submatrix) of $\boldsymbol{\mu}$ ($\boldsymbol{\Sigma}$).
- This property suggests that any marginal distribution of (multivariate) normal distribution is still a (multivariate) normal distribution.
- **Note 1:** That any marginal distribution of a random vector is a multivariate normal distribution may not suggest that the random vector obeys multivariate normal distribution.

➤ **Note 2:** A linear combination of the normal variable are not necessarily normal variable. But this conclusion holds for independent normal variables

➤ **Example 3.2.4** Let $\mathbf{x} \sim N_4(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}, \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & \sigma_{24} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} & \sigma_{34} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_{44} \end{pmatrix}$$

Then (i) $x_i \sim N(\mu_i, \sigma_{ii}), \quad i = 1, 2, 3, 4$

$$(ii) \quad \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} \mu_1 \\ \mu_4 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{14} \\ \sigma_{41} & \sigma_{44} \end{pmatrix} \right)$$

$$(iii) \quad \begin{pmatrix} x_4 \\ x_1 \\ x_3 \end{pmatrix} \sim N_3 \left(\begin{pmatrix} \mu_4 \\ \mu_1 \\ \mu_3 \end{pmatrix}, \begin{pmatrix} \sigma_{44} & \sigma_{41} & \sigma_{43} \\ \sigma_{14} & \sigma_{11} & \sigma_{13} \\ \sigma_{34} & \sigma_{31} & \sigma_{33} \end{pmatrix} \right)$$

§ 3.2 Properties of Multivariate Normal Distribution

- (5) Assume $x_1, x_2, \dots, x_n$ are mutually independent and $x_i \sim N_p(\mu_i, \Sigma_i)$, $i=1,2,\dots,n$, then for any n constant, we have

$$\sum_{i=1}^n k_i \mathbf{x}_i \sim N_p \left(\sum_{i=1}^n k_i \boldsymbol{\mu}_i, \sum_{i=1}^n k_i^2 \boldsymbol{\Sigma}_i \right)$$

- This property suggests that any linear combination of **independent** multivariate normal variable (with the same dimension) is still a multivariate normal variable.

- (6) Let $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then divide $\mathbf{x}$, $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}(>0)$ as follows:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix}_{\substack{k \\ p-k}}, \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}_{\substack{k \\ p-k}}, \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}_{\substack{k & p-k \\ k & p-k}}$$

then sub-vectors x_1 and x_2 are independent, if and only if $\Sigma_{12}=0$.

➤ The property points out that for multivariate normal variables, the fact that their sub-vectors are mutually uncorrelated is equivalent to that they are mutually independent.

➤ (7) Let $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} > 0$, then

$$(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \sim \chi^2(p)$$

➤ Example 3.2.5 Let $\mathbf{x} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$$\boldsymbol{\Sigma} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 5 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

then x_2 and x_3 are not independent, x_1 and (x_2, x_3) are independent.

➤ * (8) Omitted.

➡ * (9) Omitted.

➡ * (10) Omitted.

➡ (11) Let $x \sim N_p(\mu, \Sigma)$, $\Sigma > 0$, then divide $x, \mu, \Sigma(> 0)$ as follows:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} \begin{matrix} k \\ p-k \end{matrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix} \begin{matrix} k \\ p-k \end{matrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \begin{matrix} k & p-k \\ p-k & p-k \end{matrix}$$

The conditional distribution of x_1 given x_2 is $N_k(\boldsymbol{\mu}_{1.2}, \boldsymbol{\Sigma}_{11.2})$, where

➡ Memory
$$\boldsymbol{\mu}_{1.2} = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{x}_2 - \boldsymbol{\mu}_2) \quad (3.2.6)$$

$$\boldsymbol{\Sigma}_{11.2} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21} \quad (3.2.7)$$

➤ $\boldsymbol{\mu}_{1.2}$ and $\boldsymbol{\Sigma}_{11.2}$ are respectively the Conditional Expectation and Conditional Covariance Matrix, $\boldsymbol{\Sigma}_{11.2}$ is called Partial Covariance Matrix (偏协方差矩阵).

➡ (11) Let $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim N_p(\mu, \Sigma)$, $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} > 0$, $\mu_{1.2} = E(x_1|x_2)$,

$\Sigma_{11.2} = \text{Cov}(x_1|x_2) = E[x_1 - E(x_1|x_2)][x_1 - E(x_1|x_2)]^T$. Then, we have

$$\mu_{1.2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2) \quad (3.2.6)$$

$$\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}. \quad (3.2.7)$$

Proof: Let $A = \begin{pmatrix} I_k & -\Sigma_{12}\Sigma_{22}^{-1} \\ 0 & I_{p-k} \end{pmatrix}$ and $y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = Ax = \begin{pmatrix} x_1 - \Sigma_{12}\Sigma_{22}^{-1}x_2 \\ x_2 \end{pmatrix}$.

Then, we obtain $E(y_1) = \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2$ and

$$\text{Cov}(y) = \begin{pmatrix} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}.$$

This implies that $y_1 \sim N_k(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11.2})$. By noticing that y_1 is independent of y_2 since $\text{Cov}(y_1, y_2) = 0$, we obtain that

$$E(y_1) = E(y_1|y_2) = \mu_{1.2} - \Sigma_{12}\Sigma_{22}^{-1}x_2 = \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2.$$

This leads to $\mu_{1.2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$. By applying the definition of $\Sigma_{11.2}$, we can obtain $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$.

- This property suggests that for a multivariate normal variable, the conditional distribution of its sub-vector is still multivariate normal.
- Rewrite Equation (3.2.6):

$$E(\mathbf{x}_1|\mathbf{x}_2) = (\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2) + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{x}_2 \quad (3.2.6-1)$$

This is a regression function of a multivariate regression model for x_1 to a given variable $\mathbf{x}_2$, $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}$ is call the regression parameter matrix of x_1 to $\mathbf{x}_2$. Especially, when $k = 1$,

$$\begin{aligned} E(x_1|\mathbf{x}_2) &= \mu_1 + \boldsymbol{\sigma}'_{21}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2) \\ &= \mu_1 + \beta_2(x_2 - \mu_2) + \cdots + \beta_p(x_p - \mu_p) \\ &= \beta_0 + \beta_2x_2 + \cdots + \beta_px_p \end{aligned}$$

- This property suggests that for a multivariate normal variable, the conditional distribution of its subvector is still multivariate normal.
- **Example 3.2.7** Let $\mathbf{x} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 16 & -4 & 2 \\ -4 & 4 & -1 \\ 2 & -1 & 4 \end{pmatrix}$$

Try to give the conditional distribution of $\begin{pmatrix} x_2 - x_3 \\ x_1 \end{pmatrix}$ given $x_1 + 2x_3$.

➡ Solution: Let $y_1 = \begin{pmatrix} x_2 - x_3 \\ x_1 \end{pmatrix}$, $y_2 = x_1 + 2x_3$,

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ 1 & 0 & 2 \end{pmatrix},$$

Then,

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_2 - x_3 \\ x_1 \\ x_1 + 2x_3 \end{pmatrix} = A \times \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$E \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A \times \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$$

$$V \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = A \Sigma A' = \begin{pmatrix} 10 & -6 & -16 \\ -6 & 16 & 20 \\ -16 & 20 & 40 \end{pmatrix}$$

By using (3.2.6) and (3.2.7), conditional mean and conditional covariance matrix of y_1 for given y_2 are respectively shown as follows.

$$\mu_{1.2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2)$$

$$\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -16 \\ 20 \end{pmatrix} \frac{1}{40} (y_2 + 3) = \begin{pmatrix} -\frac{2}{5} x_1 - \frac{4}{5} x_3 + \frac{4}{5} \\ \frac{1}{2} x_1 + x_3 + 2 \frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} 10 & -6 \\ -6 & 16 \end{pmatrix} - \begin{pmatrix} -16 \\ 20 \end{pmatrix} \frac{1}{40} \begin{pmatrix} -16 & 20 \end{pmatrix} = \begin{pmatrix} 3 \frac{3}{5} & 2 \\ 2 & 6 \end{pmatrix}$$

➡ 3.12 Assume $x \sim N_3(\mu, \Sigma)$, where $\mu = (10, 4, 7)'$, and

$$\Sigma = \begin{pmatrix} 9 & -3 & -3 \\ -3 & 5 & 1 \\ -3 & 1 & 5 \end{pmatrix}$$

Problems:

(2) Find the condition distribution of $x_1 | (x_2, x_3)$ and the condition distribution of $(x_1, x_2) | x_3$.

§ 3.3 Multiple and partial correlation coefficient

- § 3.3.1 multiple correlation coefficient
- § 3.3.2 partial correlation coefficient

§ 3.3.1 multiple correlation coefficient

- (Simple) correlation coefficient measures the strength of linear relationship between two random variables y and x .
- Multiple (复) correlation coefficient measures the strength of linear relationship between one random variables y and a group of random variable $x_1, x_2, \dots, x_p$.

- Let

$$E \begin{pmatrix} y \\ x \end{pmatrix} = \begin{bmatrix} \mu_y \\ \mu_x \end{bmatrix}, V \begin{pmatrix} y \\ x \end{pmatrix} = \begin{bmatrix} \sigma_{yy} & \sigma'_{xy} \\ \sigma_{xy} & \Sigma_{xx} \end{bmatrix}, \begin{pmatrix} y \\ x \end{pmatrix} \text{ 的相关矩阵 } \begin{bmatrix} 1 & \rho'_{xy} \\ \rho_{xy} & R_{xx} \end{bmatrix},$$

- The largest correlation coefficient between y and the linear function of x is called the Multiple Correlation Coefficient (复相关系数), denoted by $\rho_{y \cdot x}$ or $\rho_{y \cdot 1, 2, \dots, p}$.

- The square of correlation coefficient between y and $l'x$ can be formulated as

$$\rho^2(y, l'x) = \frac{\text{Cov}^2(y, l'x)}{V(y)V(l'x)} = \frac{(\sigma'_{xy}l)^2}{\sigma_{yy} \cdot l' \Sigma_{xx} l}$$

(Extended Cauchy–Schwarz Inequation) if $B > 0$, then

$$\frac{(x'y)^2 \leq (x'Bx)(y'B^{-1}y)}{\sigma_{yy} \cdot l' \Sigma_{xx} l} \leq \frac{(\sigma'_{xy} \Sigma_{xx}^{-1} \sigma_{xy})(l' \Sigma_{xx} l)}{\sigma_{yy}} = \frac{(\sigma'_{xy} \Sigma_{xx}^{-1} \sigma_{xy})}{\sigma_{yy}}$$

- The multiple correlation coefficient between y and x is defined as

$$\rho_{y \cdot 1, 2, \dots, p} = \max_{l \neq 0} \rho(y, l'x) = \left(\frac{\sigma'_{xy} \Sigma_{xx}^{-1} \sigma_{xy}}{\sigma_{yy}} \right)^{1/2}$$

- when $l = \Sigma_{xx}^{-1} \sigma_{xy}$, the equality holds up.

- The multiple correlation coefficient between y and x is defined as

$$\rho_{y \cdot x} = \rho_{y \cdot 1, 2, \dots, p} = \max_{l \neq 0} \rho(y, l'x) = \left(\frac{\sigma'_{xy} \Sigma_{xx}^{-1} \sigma_{xy}}{\sigma_{yy}} \right)^{1/2}$$

- when $p = 1$, $\rho_{y \cdot x}$ reduces to simple correlation coefficient ρ_{yx} ;
- $\rho_{y \cdot 1, 2, \dots, p} = 0$ if and only if $\sigma_{xy} = 0$;
- $\rho_{y \cdot x} = \left(\frac{\sigma'_{xy} \Sigma_{xx}^{-1} \sigma_{xy}}{\sigma_{yy}} \right)^{1/2} = \sqrt{\rho'_{xy} R_{xx}^{-1} \rho_{xy}}$ (3.4.1)
- If $x_1, x_2, \dots, x_p$ are not correlated, then $R_{xx} = I$. This leads to

$$\rho_{y \cdot x}^2 = \rho'_{xy} R_{xx}^{-1} \rho_{xy} = \rho'_{xy} \rho_{xy} = \rho^2(y, x_1) + \dots + \rho^2(y, x_p)$$

- 
- **Example 3.3.1** The multiple correlation coefficient between any linear function of random variables $x_1, \dots, x_p$, denoted as $F = l_1x_1 + \dots + l_px_p$, and random variables $x_1, \dots, x_p$ is 1.

Proof: It is easy to obtain that

$$\begin{aligned}\rho_{F \cdot 1, \dots, p} &= \max_{a \neq 0} \rho(F, a_1x_1 + \dots + a_px_p) \\ &\geq \rho(F, l_1x_1 + \dots + l_px_p) \\ &= 1.\end{aligned}$$

- **Multiple Correlation Coefficient** between y and x , denoted by $\rho_{y.1,2,\dots,p}$, measures the degree of correlation between y and $x_1, x_2, \dots, x_p$

$$\rho_{y.1,2,\dots,p} = \max_{l \neq 0} \rho(y, l'x) = \left(\frac{\sigma'_{xy} \Sigma_{xx}^{-1} \sigma_{xy}}{\sigma_{yy}} \right)^{1/2} \quad (3.4.1)$$

when $l = \Sigma_{xx}^{-1} \sigma_{xy}$, the equality holds up

- It is can be proved that for any linear function of x , $g(x)$,

$$E[y - \mu_y - \sigma'_{xy} \Sigma_{xx}^{-1} (x - \mu_x)]^2 = \min_{g(x)} E[y - g(x)]^2 \quad (3.4.5)$$

Note that $\hat{y} = E(y|x) = \mu_y + \sigma'_{xy} \Sigma_{xx}^{-1} (x - \mu_x)$, **$\hat{y}$ is called as the optimal linear prediction for y based on x .**

- The only difference between $\mu_y + \sigma'_{xy} \Sigma_{xx}^{-1} (x - \mu_x)$ and $\sigma'_{xy} \Sigma_{xx}^{-1} x$ is a constant, the simple correlation coefficient between y and

$$\hat{y} = \mu_y + \sigma'_{xy} \Sigma_{xx}^{-1} (x - \mu_x) \text{ can also be the above maximum value such that}$$

$$\rho(y, \hat{y}) = \rho(y, \sigma'_{xy} \Sigma_{xx}^{-1} x) = \rho_{y.1,2,\dots,p}. \quad (3.4.6)$$

So, $\hat{y}$ is such a linear function of x to make the correlation coefficient between y and $\hat{y}$ attains the above maximum value given by (3.4.1).

Example 3.3.2: The square of multiple correlation coefficient between y and x is equivalent to the following goodness R^2 if we think there is a regression relationship, given by $\hat{y} = E(y|x) = \mu_y + \sigma'_{xy}\Sigma_{xx}^{-1}(x - \mu_x)$. That is

$$\rho_{y \cdot x}^2 = \frac{\text{Var}(\hat{y})}{\sigma_{yy}} = R^2 \rightarrow \text{Goodness of Fit (拟合优度)} \quad (3.4.11)$$

Proof: Prediction accuracy of optimal linear prediction $\hat{y}$ is defined as

$$\begin{aligned} E(y - \hat{y})^2 &= E\{y - [\mu_y + \sigma'_{xy}\Sigma_{xx}^{-1}(x - \mu_x)]\}^2 \\ (\text{def of Var}) &= \text{Var}(y - \sigma'_{xy}\Sigma_{xx}^{-1}x) = \text{Cov}(y - \sigma'_{xy}\Sigma_{xx}^{-1}x, y - \sigma'_{xy}\Sigma_{xx}^{-1}x) \\ &= \sigma_{yy}(1 - \rho_{y \cdot x}^2) \end{aligned}$$

The predicted variable y can be decomposed as

$$y = \hat{y} + y - \hat{y} = [\mu_y + \sigma'_{xy}\Sigma_{xx}^{-1}(x - \mu_x)] + y - [\mu_y + \sigma'_{xy}\Sigma_{xx}^{-1}(x - \mu_x)].$$

It is notice that

$$\text{Cov}\{x, y - [\mu_y + \sigma'_{xy}\Sigma_{xx}^{-1}(x - \mu_x)]\} = \sigma_{xy} - \text{Cov}(x, x)\Sigma_{xx}^{-1}\sigma_{xy} = 0.$$

This implies that $Cov[\hat{y}, y - \hat{y}] = 0$. Therefore, we have

$$Var(y) = Var(\hat{y}) + Var(y - \hat{y}). \quad (3.4.12)$$

The prediction accuracy of the optimal linear prediction $\hat{y}$ is

$$\begin{aligned} E(y - \hat{y})^2 &= E\{y - [\mu_y + \sigma'_{xy}\Sigma_{xx}^{-1}(x - \mu_x)]\}^2 \\ &= V(y) + V(\sigma'_{xy}\Sigma_{xx}^{-1}x) - 2Cov(y, \sigma'_{xy}\Sigma_{xx}^{-1}x) \\ &= \sigma_{yy} - \sigma'_{xy}\Sigma_{xx}^{-1}\sigma_{xy} \\ &= \sigma_{yy}(1 - \rho_{y \cdot x}^2). \end{aligned} \quad (3.4.7)$$

It is obvious that $Var(y - \hat{y}) = E(y - \hat{y})^2$ due to $E(y - \hat{y}) = 0$. This combined with (3.4.7) and (3.4.12) leads to

$$\rho_{y \cdot x}^2 = 1 - \frac{E(y - \hat{y})^2}{\sigma_{yy}} = 1 - \frac{Var(y - \hat{y})}{\sigma_{yy}} = \frac{Var(\hat{y})}{\sigma_{yy}} = R^2, \rightarrow \text{Goodness of Fit (拟合优度)}$$

which represents the proportion of variance of y that can be jointly explained by $x_1, x_2, \dots, x_p$.



Application of multiple correlation coefficients

- The multiple correlation coefficient is widely used across various fields, especially in social sciences, natural sciences, and engineering.
 - **Multiple Regression Analysis:** In multiple regression analysis, the multiple correlation coefficient is an important indicator for assessing model fit.
 - **Predictive Modeling:** By building multiple regression models, the multiple correlation coefficient can assess the model's predictive ability. A higher multiple correlation coefficient suggests better prediction accuracy.
 - **Variable Selection:** In multiple regression analysis, a subset of variables with higher multiple correlation coefficients can be selected to build a more concise and effective model.
 - **Risk Assessment:** In finance, the multiple correlation coefficient can be used to assess the risk of investment portfolios and select assets with lower correlations, thus reducing overall portfolio risk.

复相关系数-百度百科

本词条由“科普中国”科学百科词条编写与应用工作项目 审核。

复相关系数是反映一个因变量与一组自变量(两个或两个以上)之间相关程度的指标，是度量复相关程度的指标。

计算方法

编辑

复相关系数是测量一个变量与其他多个变量之间线性相关程度的指标。它不能直接测算，只能采取一定的方法进行间接测算。

测定一个变量 y 与其他多个变量 $x_1, x_2, \dots, x_k$ 之间的相关系数。 $x_1, x_2, \dots, x_k$ 不能直接测算。可以考虑构造一个关于 $x_1, x_2, \dots, x_k$ 的线性组合，通过计算该线性组合与 y 之间的简单相关系数作为变量 y 与 $x_1, x_2, \dots, x_k$ 之间的复相关系数。

具体过程如下：

第一步，用 y 对 $x_1, x_2, \dots, x_k$ 作回归，得： $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_k X_k$ 。

第二步，计算简单相关系数，即为 y 与 $x_1, x_2, \dots, x_k$ 之间的复相关系数。

复相关系数的计算公式为：
$$R = \frac{\sum (y - \bar{y})(\hat{y} - \bar{y})}{\sqrt{\sum (y - \bar{y})^2 (\hat{y} - \bar{y})^2}}$$
，Matlab里面的函数 `R=corrcoef(X, Y)` [1]。

样本复相关系数 $\xrightarrow{\text{近似}}$ $\rho(y, \hat{y}) = \frac{\text{cov}(y, \hat{y})}{\sqrt{\text{var}(y)\text{var}(\hat{y})}}$



Assignment: See the assignment file



§ 3.3.2 partial correlation coefficient

- Definition: In multivariate regression analysis, under the condition of the influence of other variables being eliminated, the correlation coefficient between two variables is called **Partial Correlation Coefficient**.
- Suppose we need to calculate the correlation between X and Y, Z denotes all of the other variables.
- The partial correlation coefficient between X and Y can be considered as the simple correlation coefficient between the residuals of the linear regressions of X and Z, R_x , and Y and Z, R_y , which is called **Pearson Correlation Coefficient**.

$$\mu_{1.2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2) \quad (3.2.6)$$

Divide $\mathbf{x}$, $\Sigma(>0)$ as follows:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} \begin{matrix} k \\ p-k \end{matrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{matrix} k \\ p-k \end{matrix}$$

- Suppose $e_i = (0, \dots, 0, 1, 0, \dots, 0)'$: $k \times 1$, where 1 is the i -th element. Then according to (3.2.6), the optimal linear prediction of x_i by $\mathbf{x}_2$ is

$$\mu_i + e_i' \Sigma_{12} \Sigma_{22}^{-1} (\mathbf{x}_2 - \mu_2), i = 1, 2, \dots, k$$

Prediction Error: $\varepsilon_i = x_i - E(x_i | \mathbf{x}_2) = x_i - [\mu_i + e_i' \Sigma_{12} \Sigma_{22}^{-1} (\mathbf{x}_2 - \mu_2)]$

The correlation coefficient between ε_i and ε_j ($i, j = 1, 2, \dots, k$) describes the linear correlation of the corresponding components of $\mathbf{x}_1$ after eliminating the influence of $\mathbf{x}_2$, which is denoted as $\rho_{ij.2} = \rho(\varepsilon_i, \varepsilon_j)$.

Suppose $\varepsilon = (\varepsilon_1, \varepsilon_2, \dots, \varepsilon_k)'$, then

$$\begin{aligned}\varepsilon &= x_1 - [\mu_1 + (e_1, e_2, \dots, e_k)' \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2)] \\ &= x_1 - [\mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2)]\end{aligned}$$

Therefore

$$\begin{aligned}V(\varepsilon) &= V(x_1 - \Sigma_{12} \Sigma_{22}^{-1} x_2) \\ &= \text{Cov}(x_1 - \Sigma_{12} \Sigma_{22}^{-1} x_2, x_1 - \Sigma_{12} \Sigma_{22}^{-1} x_2) \\ &= \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} + \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \\ &= \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} = \Sigma_{11 \cdot 2}\end{aligned}$$

is the **Partial Covariance Matrix**(偏协方差矩阵) of x_1 given x_2 ,

$$\Sigma_{11 \cdot 2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \quad (3.4.14)$$

Divide $\mathbf{x}$, $\Sigma(>0)$ as follows:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} \begin{matrix} k \\ p-k \end{matrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{matrix} k & p-k \\ p-k & k \end{matrix}$$

- $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$ is also called the conditional covariance matrix(条件协方差矩阵) of $\mathbf{x}_1$ given $\mathbf{x}_2$, denoted by $(\sigma_{ij \cdot k+1, \dots, p})$.
- $\Sigma_{11.2} = (\sigma_{ij \cdot k+1, \dots, p})$ is called partial covariance(偏协方差矩阵). It denotes the covariance of x_i and x_j after eliminating the influence of $\mathbf{x}_2 = (x_{k+1}, \dots, x_p)'$.
- **For multi-normal distribution, the conditional covariance matrix is the same as the partial covariance.**

Partial correlation coefficient of x_i and x_j given $\mathbf{x}_2$ is defined as:

$$\rho_{ij \cdot k+1, \dots, p} = \frac{\sigma_{ij \cdot k+1, \dots, p}}{\sqrt{\sigma_{ii \cdot k+1, \dots, p} \sigma_{jj \cdot k+1, \dots, p}}}, \quad 1 \leq i, j \leq k \quad (3.4.15)$$

where $\Sigma_{11 \cdot 2} = (\sigma_{ij \cdot k+1, \dots, p})$

Matlab function to compute partial correlation: `partialcorr(x,y,z)`.

- For multivariate normal variable $\mathbf{x}$, since $\Sigma_{11 \cdot 2}$ is also the conditional covariance matrix, the partial and conditional correlation coefficient have the same value. Therefore, in the meantime, $\rho_{ij \cdot k+1, \dots, p}$ measures the strength of the correlation between x_i and x_j when $x_{k+1}, \dots, x_p$ are given.

Recurrence formula of partial correlation coefficient

$$\rho_{ij \bullet k+1, \dots, p} = \frac{\rho_{ij \bullet k+2, \dots, p} - \rho_{i, k+1 \bullet k+2, \dots, p} \rho_{j, k+1 \bullet k+2, \dots, p}}{\sqrt{1 - \rho_{i, k+1 \bullet k+2, \dots, p}^2} \sqrt{1 - \rho_{j, k+1 \bullet k+2, \dots, p}^2}} \quad (3.4.16)$$

measures the strength of the correlation between x_i and x_j after removing the influence of $x_{k+1}, \dots, x_p$.

Especially

$$\rho_{12 \bullet 3} = ? \quad \rho_{12 \bullet 3} = \frac{\rho_{12} - \rho_{13} \rho_{23}}{\sqrt{1 - \rho_{13}^2} \sqrt{1 - \rho_{23}^2}}$$

$$\rho_{13 \bullet 2} = ? \quad \rho_{13 \bullet 2} = \frac{\rho_{13} - \rho_{12} \rho_{23}}{\sqrt{1 - \rho_{12}^2} \sqrt{1 - \rho_{23}^2}}$$

$$\rho_{23 \bullet 1} = ? \quad \rho_{23 \bullet 1} = \frac{\rho_{23} - \rho_{12} \rho_{13}}{\sqrt{1 - \rho_{12}^2} \sqrt{1 - \rho_{13}^2}}$$

Matlab function to compute partial correlation: `partialcorr(x,y,z)`

Practice :

3.12 Assume $x \sim N_3(\mu, \Sigma)$, and $\mu = (10, 4, 7)'$, where

$$\Sigma = \begin{pmatrix} 9 & -3 & -3 \\ -3 & 5 & 1 \\ -3 & 1 & 5 \end{pmatrix}$$

Problems:

(3) Given x_3 , find the partial correlation coefficient of x_1 and x_2 ;

(4) Find the multiple correlation coefficients x_1 and (x_2, x_3) .

Solution: (3) $-\frac{\sqrt{6}}{6}$; (4) $\frac{\sqrt{3}}{3}$.

■ Assignment :

See the assignment file

§ 3.4 Maximum Likelihood Estimation and properties of estimators

- §3.4.1 joint probability density of sample $x_1, x_2, \dots, x_n$
- §3.4.2 maximum likelihood estimation of μ and Σ
- §3.4.3 maximum likelihood estimation of correlation coefficient
- §3.4.4 properties of estimators

- $x \sim N_p(\mu, \Sigma)$, $\Sigma > 0$. Suppose $x_1, x_2, \dots, x_n$ is a **Simple Random Sample**(简单随机样本) (from now on for short **Sample**) selected from population x . Then $x_1, x_2, \dots, x_n$ are independent and identically.

➤ Let

$$X = \begin{pmatrix} \mathbf{x}'_1 \\ \mathbf{x}'_2 \\ \vdots \\ \mathbf{x}'_n \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

The above matrix is called **(Sample) Data Matrix**(数据矩阵) or **Observed Value Matrix**(观测值矩阵).

Introduction

点估计的极大似然估计法(maximum likelihood) (p13)

引例:

我与一位猎人一起外出打猎，一只野兔从前方窜过，只听得一声枪响，野兔应声倒下

问 是谁打中的呢？

答：极有可能是猎人。

显然：候选人就两个，我和猎人。若选择我，则事件野兔打中的发生概率为0.01%；若选择猎人，则事件发生的概率为99%。而事件已经发生，因此选择猎人更为合理。

Introduction

➤ Basic idea for maximum likelihood estimation :

- (For parameter space). Assume that the population contains many parameters θ to be estimated, which may have many possible values. For example, we know that the rabbit has been over.
- (For likelihood idea) Given the sample observations value, one need choose the value that makes the probability of the observations to be the maximum from all possible values of θ as the estimate of θ , which is called the maximum likelihood estimate.
- Maximum likelihood, as the name implies, is the meaning that seems to be the most possible.

§3.4.1 joint probability density of sample $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$

- **Maximum Likelihood Estimation(MLE)**(极大似然估计) is obtained by maximizing **Likelihood Function**(似然函数).
- Likelihood function take the forms of sample joint probability density function $f(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$, denoted by $L(\boldsymbol{\mu}, \boldsymbol{\Sigma})$:

$$\begin{aligned} L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) &= f(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = \prod_{i=1}^n f(\mathbf{x}_i) \\ &= \prod_{i=1}^n (2\pi)^{-p/2} |\boldsymbol{\Sigma}|^{-1/2} \exp\left[-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})\right] \\ &= \left[(2\pi)^p |\boldsymbol{\Sigma}|\right]^{-n/2} \exp\left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})\right] \quad (3.4.4) \end{aligned}$$

§3.4.2 maximum likelihood estimation of μ and Σ

➤ Univariate:

$$L(\hat{\mu}, \hat{\sigma}^2) = \max_{\mu, \sigma^2} L(\mu, \sigma^2)$$

$$\hat{\mu} = \bar{x}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

➤ Multivariate:

$$L(\hat{\mu}, \hat{\Sigma}) = \max_{\mu, \Sigma} L(\mu, \Sigma)$$

$$\hat{\mu} = \bar{x}, \quad \hat{\Sigma} = \frac{1}{n} A$$

➤ $\bar{x}$ is called Sample Mean Vector(样本均值向量)

➤ $A = \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})'$ is called Sample Dispersion Matrix(样本离差矩阵).

In order to figure out the value of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ when (3.4.4) takes the extreme value, we take the logarithm of both sides of (3.3.3).

$$\ln L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{1}{2} pn \ln(2\pi) - \frac{n}{2} \ln |\boldsymbol{\Sigma}| - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})$$

Since log function is strictly monotonously increasing, the estimator of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ can be obtained through maximum of $\ln L(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Note: As matrix algebra theory, for real symmetric matrix $\mathbf{A}$,

$$\frac{\partial(\mathbf{X}'\mathbf{A}\mathbf{X})}{\partial \mathbf{X}} = 2\mathbf{A}\mathbf{X}, \quad \frac{\partial(\mathbf{X}'\mathbf{A}\mathbf{X})}{\partial \mathbf{A}} = \mathbf{X}\mathbf{X}', \quad \frac{\partial \ln |\mathbf{A}|}{\partial \mathbf{A}} = \mathbf{A}^{-1}.$$

We take partial derivative of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, respectively, with respect to log likelihood function, then

$$\begin{cases} \frac{\partial \ln L(\boldsymbol{\mu}, \boldsymbol{\Sigma})}{\partial \boldsymbol{\mu}} = \sum_{i=1}^n \boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) = 0 \\ \frac{\partial \ln L(\boldsymbol{\mu}, \boldsymbol{\Sigma})}{\partial \boldsymbol{\Sigma}} = -\frac{n}{2} \boldsymbol{\Sigma}^{-1} + \frac{1}{2} (\boldsymbol{\Sigma}^{-1})^2 \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})' = 0 \end{cases} \quad (2.18)$$

From (2.18) we can obtain the MLEs

$$\begin{cases} \hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = \bar{\mathbf{x}} \\ \hat{\boldsymbol{\Sigma}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})' = \frac{1}{n} A \longrightarrow \text{Sample dispersion matrix} \end{cases}$$

Therefore, the maximum likelihood estimator of the mean vector of multivariate normal population, $\boldsymbol{\mu}$, is sample mean vector. The maximum likelihood estimator of covariance matrix $\boldsymbol{\Sigma}$ is sample covariance matrix.

$$\begin{aligned} & \frac{\partial \left\{ -\frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \right\}}{\partial \boldsymbol{\Sigma}} \\ &= -\frac{1}{2} \frac{\partial \text{tr} \{ (\mathbf{x}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \}}{\partial \boldsymbol{\Sigma}} \\ &= -\frac{1}{2} \frac{\partial \text{tr} \{ \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})' \}}{\partial \boldsymbol{\Sigma}} \\ &= -\frac{1}{2} \frac{\partial \text{tr} \{ \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})' \}}{\partial \boldsymbol{\Sigma}} \\ &= \frac{1}{2} \boldsymbol{\Sigma}^{-2} (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})' \end{aligned}$$

Properties of maximum likelihood estimation(MLE)

- **Invariance principle of MLE.** If $\hat{\theta}$ is the MLE of θ , for any function $\tau(\theta)$ which is a one to one mapping such that $\tau^{-1}(\theta)$ exists, then the MLE of $\tau(\theta)$ is $\tau(\hat{\theta})$.
- Based on this property, we next introduce the definitions of MLE of correlation coefficients.

§3.4.3 maximum likelihood estimation of correlation coefficient

- I) simple correlation coefficient
- II) multiple correlation coefficient
- III) partial correlation coefficient

I) simple correlation coefficient

➤ MLE of correlation coefficient ρ_{ij} :

$$r_{ij} = \frac{\hat{\sigma}_{ij}}{\sqrt{\hat{\sigma}_{ii}} \sqrt{\hat{\sigma}_{jj}}} = \frac{\sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)}{\sqrt{\sum_{k=1}^n (x_{ki} - \bar{x}_i)^2} \sqrt{\sum_{k=1}^n (x_{kj} - \bar{x}_j)^2}}$$

where $\hat{\Sigma} = (\hat{\sigma}_{ij})$, $\bar{\mathbf{x}} = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_p)'$, r_{ij} is called **Sample Correlation Parameters** (样本相关系数) and $\hat{\mathbf{R}} = (r_{ij})$ is called **Sample Correlation Matrix** (样本相关矩阵).

I) simple correlation coefficient

- MLE of correlation coefficient ρ_{ij} can also be defined as:

$$r_{ij} = \frac{s_{ij}}{\sqrt{s_{ii}} \sqrt{s_{jj}}} = \frac{\frac{1}{n-1} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)}{\sqrt{\frac{1}{n-1} \sum_{k=1}^n (x_{ki} - \bar{x}_i)^2} \sqrt{\frac{1}{n-1} \sum_{k=1}^n (x_{kj} - \bar{x}_j)^2}}$$

where $\bar{\mathbf{x}} = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_p)'$, $\mathbf{S} = \frac{1}{n-1} \mathbf{A} = (s_{ij})$. $\mathbf{S}$ is called **Sample Covariance Matrix**(样本协方差矩阵), r_{ij} is called **Sample Correlation Parameters**(样本相关系数) and $\mathbf{R} = (r_{ij})$ is called **Sample Correlation Matrix**(样本相关矩阵)

II) multiple correlation coefficient

➡ Divide $\mathbf{x}$, $\Sigma (> 0)$ as follows:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \mathbf{x}_2 \end{pmatrix} \begin{matrix} 1 \\ p-1 \end{matrix}, \quad \Sigma = \begin{pmatrix} \sigma_{11} & \boldsymbol{\sigma}'_{21} \\ \boldsymbol{\sigma}_{21} & \Sigma_{22} \end{pmatrix} \begin{matrix} 1 \\ p-1 \end{matrix}, \quad \mathbf{S} = \begin{pmatrix} s_{11} & \mathbf{s}'_{21} \\ \mathbf{s}_{21} & \mathbf{S}_{22} \end{pmatrix} \begin{matrix} 1 \\ p-1 \end{matrix}$$

Then the MLE of multiple correlation coefficient $\rho_{1 \cdot 2, \dots, p}$, $r_{1 \cdot 2, \dots, p}$ is called **Sample Multiple Correlation Coefficient** (样本复相关系数).

$$\rho_{1 \cdot 2, \dots, p} = \left(\frac{\boldsymbol{\sigma}'_{21} \Sigma_{22}^{-1} \boldsymbol{\sigma}_{21}}{\sigma_{11}} \right)^{1/2}, \quad r_{1 \cdot 2, \dots, p} = \left(\frac{\mathbf{s}'_{21} \mathbf{S}_{22}^{-1} \mathbf{s}_{21}}{s_{11}} \right)^{1/2}$$

III) partial correlation coefficient

➡ Divide $\mathbf{x}$, $\Sigma (>0)$ as follows:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} \begin{matrix} k \\ p-k \end{matrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{matrix} k \\ p-k \end{matrix}, \quad \mathbf{S} = \begin{pmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{pmatrix} \begin{matrix} k \\ p-k \end{matrix}$$

The partial correlation coefficient

$$\rho_{ij \bullet k+1, \dots, p} = \frac{\sigma_{ij \bullet k+1, \dots, p}}{\sqrt{\sigma_{ii \bullet k+1, \dots, p} \sigma_{jj \bullet k+1, \dots, p}}}, \quad 1 \leq i, j \leq k$$

III) partial correlation coefficient

Then the MLE of partial correlation coefficient $\rho_{ij \cdot k+1, \dots, p}$, $r_{ij \cdot k+1, \dots, p}$ is called **Sample Partial Correlation Coefficient** (样本偏相关系数)

$$\Sigma_{11 \cdot 2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} = (\sigma_{ij \cdot k+1, \dots, p})$$

$$S_{11 \cdot 2} = S_{11} - S_{12} S_{22}^{-1} S_{21} = (s_{ij \cdot k+1, \dots, p})$$

$$r_{ij \cdot k+1, \dots, p} = \frac{s_{ij \cdot k+1, \dots, p}}{\sqrt{s_{ii \cdot k+1, \dots, p} s_{jj \cdot k+1, \dots, p}}}, \quad 1 \leq i, j \leq k$$



➡ Assignment :

See the assignment file



§3.4.4 properties of estimators

- 1. Unbiasness
- 2. Effectiveness
- 3. Consistence
- 4. Sufficiency

1.Unbiasness

- Suppose $\hat{\theta}$ is an estimator of unknown parameter θ (vector or matrix). If $E(\hat{\theta}) = \theta$, then estimator $\hat{\theta}$ is called **unbiased estimation**(无偏估计) of θ , otherwise it is biased.

Sample mean $\bar{x}$ is the unbiased estimation of population mean μ .

$$E(\bar{x}) = \mu$$

The MLE $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})'$ is not the unbiased estimation of Σ .

$$E(\hat{\Sigma}) = \frac{n-1}{n} \Sigma$$

- If the estimator is slightly corrected as

$$S = \frac{n\hat{\Sigma}}{n-1} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})',$$

then S is a unbiased estimator of Σ . That is $E(S) = \Sigma$.

$\hat{\Sigma}$ is not the unbiased estimate of Σ :

$$\begin{aligned} E\hat{\Sigma} &= E\left(\frac{1}{n}A\right) = \frac{1}{n}E\left[\sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})'\right] \\ &= \frac{1}{n}E\left[\sum_{i=1}^n (X_i - \mu)(X_i - \mu)' - n(\bar{X} - \mu)(\bar{X} - \mu)'\right] \\ &= \frac{1}{n}\left(\sum_{i=1}^n \text{var } X_i - n \text{var } \bar{X}\right) \\ &= \frac{n-1}{n}\Sigma \end{aligned}$$

2 Effectiveness

- Suppose there exist two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ of θ . If

$$V(\hat{\theta}_1) \leq V(\hat{\theta}_2)$$

holds for all the $\theta \in \Theta$, then $\hat{\theta}_1$ has higher estimation efficiency than $\hat{\theta}_2$. For short, $\hat{\theta}_1$ is more effective than $\hat{\theta}_2$.

- The uniformly minimum variance unbiased estimation (UMVUE, 一致最小方差无偏估计): for any one of θ 's unbiased estimator $\tilde{\theta}$, the unbiased $\hat{\theta}$ satisfies

$$V(\hat{\theta}) \leq V(\tilde{\theta}), \theta \in \Theta.$$

If θ is a **parameter vector**, then $\hat{\theta}$ is the uniformly most powerful unbiased estimation (UMPUE, 一致最优无偏估计) of θ .

3 Consistence (相合性)

- If the estimator $\hat{\theta}_n$ of unknown parameters θ infinitely close to true value (convergence in probability, $\hat{\theta}_n \xrightarrow{p} \theta$) with the increase of sample size n , $\hat{\theta}_n$ is called the consistent estimator of θ .
- $\bar{x}$ and $\hat{\Sigma}$ are respectively the consistent estimators of μ and Σ since $\hat{\Sigma} = \frac{n-1}{n} S$ and $E(S) = \Sigma$.

§ 3.5 Sampling distribution of $\bar{x}$ and s^2

- Draw a sample $x_1, x_2, \dots, x_n$ from the population $N(\mu, \sigma^2)$.

Let the sample mean and the sample variance be $\bar{x}$, s^2 whose definitions are given by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Then, the sampling distribution of $\bar{x}$ and s^2 are as follows:

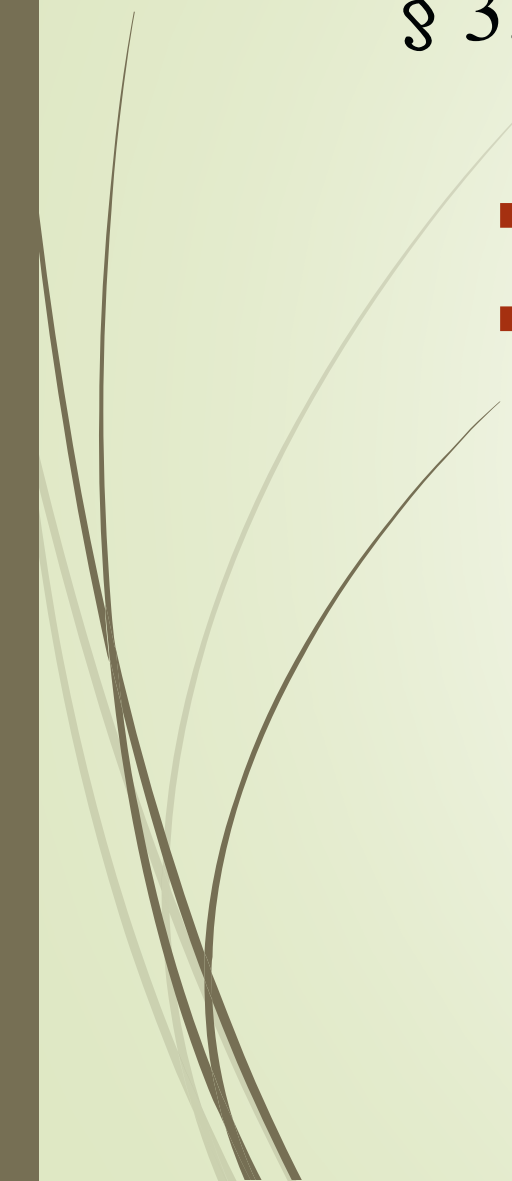
(i) $\bar{x}$ and s^2 are independent.

(ii) $\bar{x} \sim N\left(0, \frac{1}{n} \sigma^2\right)$ and $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$.

Proofs can be solved by Exercise 3.15



§ 3.5 Sampling Distribution about $\bar{x}$ and $(n - 1) S$

- § 3.5.1 Sampling Distribution about $\bar{x}$
 - § 3.5.2 Sampling Distribution about $(n - 1) S$
- 

§ 3.5.1 Sample Distribution about $\bar{\mathbf{x}}$

➤ 1. Normal Population

Suppose $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} > 0$, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ is a sample from population $\mathbf{x}$, then

$$\bar{\mathbf{x}} \sim N_p\left(\boldsymbol{\mu}, \frac{1}{n} \boldsymbol{\Sigma}\right) \quad (3.5.1)$$

➤ 2. Non-normal Population (Multivariate Central-Limit Theorem)

Suppose $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ is a sample from population $\mathbf{x}$, $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ exist. Then when n is large and p fixed, (3.5.1) holds, i.e.,

$$\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \sim N_p(0, \boldsymbol{\Sigma}) \quad (3.5.2)$$

Theorem 3.5.1. Suppose $\bar{X}$ and $\mathbf{S} = \frac{1}{n-1} \mathbf{A}$ are respectively sample mean and sample covariance from the population distribution $N_p(\mu, \Sigma)$, where $\mathbf{A} = \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})'$. Then, we have the following conclusions.

i) $\bar{X} \sim N_p\left(\mu, \frac{1}{n} \Sigma\right)$;

ii) $\bar{X}$ and $\mathbf{S}$ are mutually independent.

iii) Dispersion matrix $\mathbf{A} = (n-1)\mathbf{S}$ can be written as

$$(n-1)\mathbf{S} = \sum_{i=1}^{n-1} \mathbf{Z}_i \mathbf{Z}_i', \longrightarrow \text{Distributed as Wishart distribution with df } n-1$$

where $\mathbf{Z}_1, \dots, \mathbf{Z}_{n-1} \xrightarrow{i.i.d} N(0, \Sigma)$.

➤ Before introducing Wishart distribution, we first introduce how to define a distribution of random matrix.

§3.5.2 The Sample Distribution about $(n - 1)S$ --Wishart Distribution

- Let random matrix $X = (x_1, x_2, \dots, x_q) = (x_{ij}): p \times q$. we make a long vector from the column vectors of X one by one, denoted by $vec(X)$, that is

$$vec(X) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_q \end{bmatrix},$$

abbreviated as straightening (拉直) operation. If X is a symmetric matrix of order p with $x_{ij} = x_{ji}$, we just need to take the lower triangular part of X to form a reduced long vector, denoted by

$$vech(X) = (x_{11}, \dots, x_{p1}, x_{22}, \dots, x_{p2}, \dots, x_{p-1,p-1}, x_{p,p-1}, x_{p,p})'.$$

- The distribution of random matrix X is the distribution of $vec(X)$ or $vech(X)$.

§3.5.2 Sample Distribution about $(n - 1)S$ ----- Wishart Distribution

- Definition: Suppose $X_i \sim N_p(0, \Sigma) (i = 1, 2, \dots, n)$ are mutually independent. Then, the random matrix made up of X_i

$$W_{p \times p} = \sum_{i=1}^n X_i X_i'$$

is called as Wishart distribution with degree of n , denoted by $W_p(n, \Sigma)$.

- When $p = 1, \Sigma = \sigma^2$, $W = \sum_{i=1}^n x_i^2 \sim \sigma^2 \chi_n^2$
- Wishart distribution is the multivariate extension of χ^2 distribution under p -dimensional normal condition.

§3.5.2 Sample Distribution about $(n - 1)S$ ----- Wishart Distribution

- The Wishart distribution of a positive definite matrix has the pdf

$$f(W|n, \Sigma) = \frac{|W|^{\frac{1}{2}(n-p-1)} e^{-\frac{1}{2}\text{tr}(\Sigma^{-1}W)}}{2^{np/2} \Gamma_p\left(\frac{n}{2}\right) |\Sigma|^{n/2}},$$

where p is the dimension of W , $\Gamma_p(\cdot)$ is a multivariate generalization of the Gamma function $\Gamma(\cdot)$ with $\Gamma_p\left(\frac{n}{2}\right) = \pi^{p(p-1)/4} \prod_{j=1}^p \Gamma\left(\frac{n+1-j}{2}\right)$.

- The Wishart distribution has mean $n\Sigma$ and variance $\text{Var}(W_{ij}) = n(\sigma_{ij}^2 + \sigma_{ii}\sigma_{jj})$, where $\Sigma = (\sigma_{ij}), i, j = 1, 2, \dots, p$.
- It is not necessary to memorize the pdf of Wishart distribution.
- In Bayesian statistics, in the context of the multivariate normal distribution, the Wishart distribution is often used as the conjugate prior to the precision matrix $\Omega = \Sigma^{-1}$, where Σ is the covariance matrix.

§3.5.2 To generate random numbers from the Wishart distribution

Matlab

 Copy code

```
1 n = 5; % 自由度
2 C = eye(3); % 尺度矩阵为单位矩阵
3 R = wishrnd(C, n);
```

python

```
import numpy as np
from scipy.stats import wishart

# 设置自由度df和尺度矩阵scale
df = 5 # 自由度
scale = np.array([[1, 0.5], [0.5, 1]]) # 尺度矩阵

# 生成Wishart分布的随机数
random_matrix = wishart.rvs(df, scale)

print("生成的Wishart分布随机矩阵: ")
print(random_matrix)
```

需要安装Statistics and
Machine Learning Toolbox

生成的Wishart分布随机矩阵:

```
[[2.20895948 0.27661076]
 [0.27661076 1.82342821]]
```

Properties of Wishart distribution:

1. (Additivity) If $W_i \sim W_p(n_i, \Sigma)$, $i = 1, 2, \dots, k$ are mutually independent, then,

$$\sum_{i=1}^n W_i \sim W_p\left(\sum_{i=1}^k n_i, \Sigma\right).$$

2. If a random matrix $W_{p \times p} \sim W_p(n, \Sigma)$, $C_{p \times p}$ is a non singular matrix, then

$$CWC' \sim W_p(n, C\Sigma C').$$

3. If $X_i \sim N_p(\mu, \Sigma)$, $i = 1, 2, \dots, n$, are mutually independent, then the sample dispersion matrix

$$(n-1)S = \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' \sim W_p(n-1, \Sigma)$$

$$\text{with } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Exercises:

1. Assume $X_1, X_2, \dots, X_n$ are the random sample from $N_p(\mu, \Sigma)$, $C_i \geq 0$, $i = 1, 2, \dots, n$, $\sum_{i=1}^n C_i = 1$. Let $Z = \sum_{i=1}^n C_i X_i$. Please prove that
 - (1) Z is an unbiased estimator of μ ;
 - (2) $Z \sim N_p(\mu, C' C \Sigma)$, where $C = (C_1, \dots, C_n)'$;
 - (3) When $C = \frac{1}{n} \mathbf{1}_n$, the covariance matrix of Z attains the minimum in the sense of a nonnegative definite.
2. Let $X_1, X_2, \dots, X_n$ be a random sample from $N_5(\mu, \Sigma)$, try to derive the distribution of the following statistic:
 - (1) The distribution of $(X_1 - \mu)' \Sigma^{-1} (X_1 - \mu)$; $\chi^2(5)$
 - (2) The distribution of $\sqrt{n}(\bar{X} - \mu)$. $N_5(0, \Sigma)$
 - (3) The distribution of $(n-1)S$, where $S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})'$. $W_5(n-1, \Sigma)$